WEEK 8 OF 15 · STAGE 2: GCAM IN DEPTH

The energy system in GCAM

Change one cost and follow the ripples

New R skill Comparing scenarios: differences from reference with a join; rchart for standard GCAM charts

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	Week 4 Economics primer	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	**Week 8 Energy in GCAM**	Week 9 Land, water and bioenergy	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	Week 12 Why models disagree	Week 13 Uncertainty	Week 14 Frontiers and limits	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Trace energy from resources through power plants and refineries to final demand
2. Calculate a technology's levelised cost from its three parts
3. Explain why new investment changes fast but the whole fleet changes slowly
4. Run a scenario with an add-on file and compute differences from Reference in R

? **Big question** If solar power becomes much cheaper, how fast does the electricity mix change, and why not overnight?

Solar gets cheaper tomorrow. What happens?

Suppose new solar panels suddenly cost the same as coal. Your region's power plants are mostly coal and gas today.



How long until most of the electricity comes from solar: 5 years, 20 years, or longer?

AN EVERYDAY PICTURE

Energy moves along three conveyor belts



Think of an engine hall with three belts. The first carries raw resources: oil barrels, coal, sunlight and wind. The second runs through power stations, refineries and hydrogen plants. The third delivers to homes, lorries and steel mills. Every handover between belts is a market with a price.

What it means for models

1. Belt 1: resources (primary energy)
2. Belt 2: transformation (electricity, fuels, hydrogen)
3. Belt 3: final demand in buildings, transport and industry

HOW IT WORKS

Demand for services pulls energy through the system

1

Resources

Oil, gas, coal, uranium, wind, sun, biomass

2

Transformation

Power plants, refineries, hydrogen production

3

Final energy

Electricity, liquid fuels, gas, hydrogen delivered

4

Services

Warm homes, kilometres travelled, tonnes of steel

Fossil resources are graded: the cheap grades are used first, so extraction cost rises over time

Worked example: what does gas power cost?

Problem Levelised cost = non-energy cost + fuel price ÷ efficiency + carbon price × emissions per MWh. Gas plant, 50% efficient, carbon price $50/t.

1. Non-energy cost (building and running the plant): $20 per MWh
2. Fuel: gas at $25 per MWh ÷ 0.5 efficiency = $50
3. Emissions: 0.2 t per MWh of gas ÷ 0.5 = 0.4 t per MWh of electricity
4. Carbon cost: 0.4 × $50 = $20
5. Total: 20 + 50 + 20 = $90 per MWh

Answer **$90 per MWh, the same as week 4's gas cost with a $50 carbon price**

SIDE BY SIDE

Three parts of cost, three different sources

Part of cost	Where it comes from	Example change
Non-energy cost	Input file, per technology and year	CheapSolar lowers it
Fuel ÷ efficiency	The fuel market, solved by GCAM	A gas shortage raises it
Carbon price × emissions	The policy you add	CarbonPrice in week 10

Future costs and efficiencies are assumptions written in input files before the run

Quick check

Coal plant: non-energy $25/MWh, coal $14/MWh, 40% efficient, 0.36 tCO2 per MWh of coal. Cost at $50/t?

A $75

B $85

C $105

D $120

Quick check: answer

Coal plant: non-energy $25/MWh, coal $14/MWh, 40% efficient, 0.36 tCO2 per MWh of coal. Cost at $50/t?

A $75

B $85

C **$105** ✓

D $120

Worked example: CheapSolar's new investment

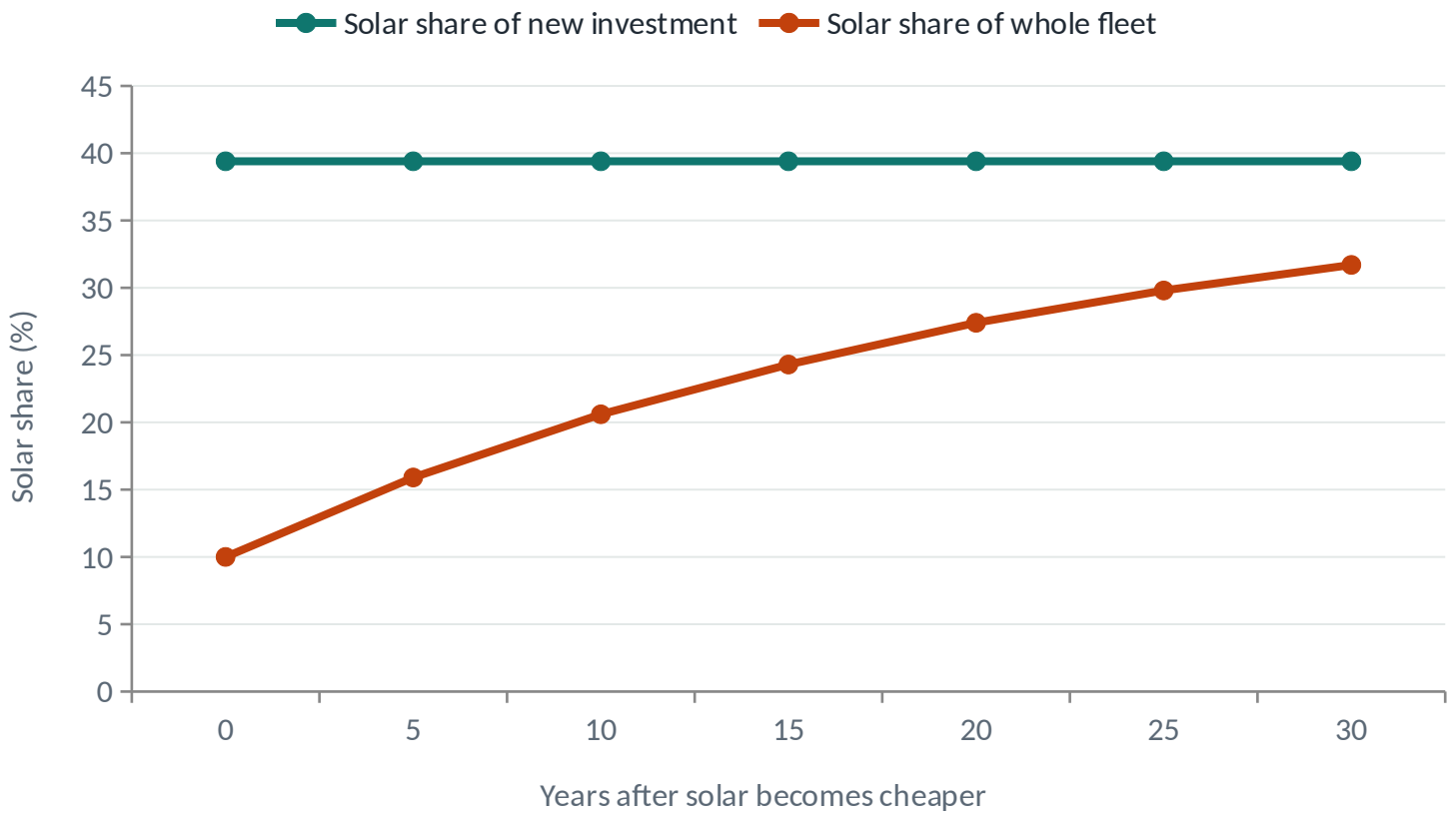
Problem Before: coal $60, gas $70, solar $80, exponent −4 gave 54 / 29 / 17. CheapSolar makes solar cost $60.

1. Coal: 60^−4 = 7.72 × 10^−8
2. Solar now also 60^−4 = 7.72 × 10^−8
3. Gas: 70^−4 = 4.16 × 10^−8
4. Sum = 7.72 + 7.72 + 4.16 = 19.6 (× 10^−8)
5. Shares: 7.72 ÷ 19.6 = 39%, gas 4.16 ÷ 19.6 = 21%

Answer **Coal 39%, gas 21%, solar 39% of new investment**

PICTURE THE NUMBERS

New investment jumps; the whole fleet creeps



Illustrative fleet: solar starts at 10%, 20% replaced per period

Read it like this

1. Each 5 years, 20% of the fleet retires and is replaced
2. Next fleet share = 0.8 × old + 0.2 × 39.4%
3. After 30 years solar is still under a third

SIDE BY SIDE

Where future cost assumptions come from: learning curves

Installed capacity	Cost index (20% learning rate)
1× today	100
2× today	80
4× today	64
8× today	51
16× today	41

Each doubling of capacity cuts cost by 20%. Illustrative learning rate.

KEY IDEA

How to build a CheapSolar scenario

1

Copy the configuration

cp configuration_ref.xml config_cheapsolar.xml. Never edit the original.

2

Add one file, last

List solar_adv.xml at the end of the inputs, just before the solver line. Later files win.

3

Rename the scenario

Change scenarioName from Reference to CheapSolar, or your results overwrite each other's labels.

4

Compare, don't eyeball

Subtract Reference from CheapSolar year by year in R.

WATCH OUT

Common mix-ups

Not quite	Better
GCAM shows that solar will get cheaper	You told it so. Future costs are inputs; cheap solar winning partly restates your assumption.
Comparing two scenarios' totals by eye	Always compute the difference from Reference, year by year.
A cheaper technology takes over within one period	Only new investment shifts. Old plants keep running until they retire.

Words to add to the class glossary

Levelised cost The average cost of one unit of output (for example one MWh) over a plant's whole life, including building, fuel and any carbon price.

Efficiency How much useful output you get per unit of fuel. A 50% efficient plant burns 2 MWh of gas for every 1 MWh of electricity.

Vintage The year a plant was built. GCAM keeps track of vintages, so old plants keep running until they retire.

Add-on file An extra input file listed last in the configuration. It overrides earlier data, so you can change one assumption without editing the originals.

Learning rate The percentage a technology's cost falls each time its total installed capacity doubles. A common way to build future cost assumptions.

Lab 8: CheapSolar versus Reference

Build and start a CheapSolar run with an add-on file, then add the instructor's CheapSolar results to your project and compute differences from Reference.

Script: labs/lab8.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

Build the CheapSolar run (Terminal tab)

Copy the configuration, make the two edits shown in the snippet file, and start the run.

```
cd ~/gcam/exe
tail -n 3 logs/main_log.txt   # did lab 7 finish?
cp configuration_ref.xml config_cheapsolar.xml
# edit config_cheapsolar.xml (two changes), then:
nohup ./gcam.exe -C config_cheapsolar.xml &
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Add CheapSolar to your project

Load last week's project and add the new scenario. Queries already stored for Reference are not run again.

```
library(tidyverse)
library(rgcam)
db_dir <- "/course/gcam/output"  # from instructor
conn <- localDBConn(db_dir, "database_basexdb")
prj <- loadProject("output/gcam_course.dat")
prj <- addScenario(conn, prj, "CheapSolar",
  "gcam-files/course_queries.xml")
elec <- getQuery(prj, "elec gen by subsector") |>
  filter(region == "Japan")   # your region
```

Take differences from Reference

Make a small Reference table, join it to CheapSolar by fuel and year, and subtract.

```
ref <- elec |>
  filter(scenario == "Reference") |>
  select(subsector, year, ref = value)
dif <- elec |>
  filter(scenario == "CheapSolar") |>
  left_join(ref, by = c("subsector", "year")) |>
  mutate(change = value - ref)
ggplot(dif, aes(year, change, fill = subsector)) +
  geom_col()
```

Let rchart draw the standard charts

rchart wants columns scenario, region, year, value, class and param. With scenRef set, it also draws difference charts.

```
library(rchart)
rc <- elec |>
  select(scenario, region, year, value,
         class = subsector) |>
  mutate(param = "Electricity")
charts <- chart(rc, scenRef = "Reference",
                save = FALSE)
names(charts)
charts$chart_class_diff_absolute
```

Did it work?

You should see

- dif has one row per fuel group and year
- Changes are zero before the first future period
- Solar's change is positive; coal and gas lose a little each

Worksheet questions

1. By how many units does solar generation change in your region in 2030 and in 2050? Why is 2050 larger?
2. Does total electricity change? Explain using the electricity price.
3. Repeat the difference for CO2 emissions. Is cheap solar alone a climate policy?

RECAP

Take-home messages

1. Energy flows from resources through transformation to final demand; each handover is a market
2. Levelised cost = non-energy cost + fuel ÷ efficiency + carbon price × emissions
3. Future costs are assumptions you set; learning curves are one way to choose them
4. A cost cut shifts new investment fast, but the fleet turns over slowly
5. Always report results as differences from Reference

Next: week 9

Week 9 turns to the other half of GCAM: farms, forests and water, and how energy, land and water pull on each other.

Practise

Interactive course, week 8: try the learning explorer and the 5-question quiz.

Readings: rchart package guide; GCAM v8.2 documentation: Energy Supply, Energy Demand, Technology Representation